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Sensitivity of Causal Effect Estimates Under Assumed Gaussian Noise to the True Latent Noise Distribution

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Abstract

Estimating causal effects from observational data is frequently challenged by the presence of unobserved confounders. The ρ-GNF estimator addresses this challenge by using deep generative models to perform sensitivity analysis, parametrizing the latent confounding structure using a Gaussian copula. However, the reliability of this estimator when the true underlying latent distribution deviates from the Gaussian copula assumption remains a critical open question.

This thesis systematically evaluates the robustness of the ρ-GNF to misspecified latent structures. Utilizing a controlled simulation based strategy, we tested the model against data generating processes exhibiting non-linear (quadratic) and asymmetric tail-dependent (Clayton copula) latent structures. The analysis focused on whether the estimator could determine an "effective" sensitivity parameter that recovers the true Average Causal Effect (ACE) despite the distributional mismatch.

Our results demonstrate that the ρ-GNF is remarkably robust to structural misspecification. In both non-linear and tail-dependent scenarios, the model consistently yielded a specific sensitivity parameter that compensated for the bias, aligning the estimated ACE with the ground truth. Furthermore, the estimator exhibited high stability and monotonicity across repeated experiments. These findings suggest that the Gaussian copula serves as a flexible functional proxy for complex dependency structures, extending the practical applicability of the ρ-GNF beyond strictly Gaussian copula environments.

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Contents

Abstract **iii**

Acknowledgments **iv**

Contents **v**

List of Figures **vi**

1 Introduction **1**
1.1 Motivation 1
1.2 Aim and Research Question 2

2 Theoretical Background and Related Work **3**
2.1 Theoretical Background 3
2.2 Related Work 6

3 Method **8**
3.1 Experimental Design Strategy 8
3.2 Data Generating Processes 9
3.3 Implementation and Evaluation 11

4 Results **14**
4.1 Gaussian Copula DGP 14
4.2 Non-Gaussian DGP 17
4.3 Clayton Copula DGP 20

5 Discussion **24**
5.1 Results 24
5.2 Methodological Evaluation 25
5.3 Ethical Considerations 26
5.4 Future Work 27

6 Conclusion **28**

Bibliography **30**

List of Figures

2.1 DAG illustrating the causal structure with copula-based confounding. The treatment T and outcome Y are influenced by their respective unobserved latent terms ϵ_T and ϵ_Y. The dependency structure between these latent terms is modeled by the copula C_ρ, representing the unobserved confounding. 6

4.1 Estimated ρ-curves for the Gaussian DGP (Baseline) with fixed true correlation $\rho_{true} = 0.7$. The graphs show the true causal effect as follows: (a) $\alpha = -2.0$, (b) $\alpha = 1.0$, and (c) $\alpha = 2.0$. The blue curve is the mean estimated ACE over 20 independent runs and the shaded region indicates the standard deviation. The horizontal red dashed line indicates the ground truth ACE while the vertical green dashed line indicates the true confounding level $\rho = 0.7$. 15

4.2 Estimated ρ-curves for the Gaussian DGP (Baseline) under varying confounding strengths. The graphs show the different true confounding levels as follows: (a) $\rho = -0.7$, (b) $\rho = 0$, and (c) $\rho = 0.7$. In all cases, the true causal effect is fixed at $\alpha = 1.0$. The blue curve represent the mean estimated ACE over 20 independent runs and the shaded region indicates the standard deviation. The horizontal red dashed line marks the ground truth ACE while the vertical green dashed line marks the respective true confounding level (ρ_{true}) 17

4.3 Estimated ρ-curves for the Quadratic scenario ($\beta = 0.5$). The graphs correspond to true causal effects: (a) $\alpha = -2.0$, (b) $\alpha = 1.0$, and (c) $\alpha = 2.0$. Since the true dependency is non-Gaussian, there is no vertical marker for ρ_{true}. The intersection of the blue curve with the red true ACE line indicates the "effective ρ" value where the estimate aligns with the ground truth. 18

4.4 Estimated ρ-curves for the Quadratic scenario with fixed $\alpha = 1.0$ and varying non-linear strengths. (b) Represents the control case where the dependency is linear ($\beta = 0$). (a) and (c) represent misspecified quadratic dependencies. The intersection of the blue curve with the red true ACE line denotes the effective sensitivity parameter. 20

4.5 Estimated ρ-curves for the Archimedean (Clayton) scenario ($\theta = 2.0$). The graphs correspond to true causal effects: (a) $\alpha = -2.0$, (b) $\alpha = 1.0$, and (c) $\alpha = 2.0$. Since the true dependency involves tail dependence (non-Gaussian), there is no vertical marker for ρ_{true}. The intersection of the blue curve with the red true ACE line denotes the effective sensitivity parameter. 21

4.6 Estimated ρ-curves for the Archimedean (Clayton) scenario with fixed $\alpha = 1.0$ and varying tail dependence strengths. The graphs correspond to: (a) Weak dependence $\theta = 0.1$, (b) Moderate dependence $\theta = 2.0$, and (c) Strong dependence $\theta = 5.0$. The intersection of the blue curve with the red true ACE line indicates the effective sensitivity parameter required to approximate the bias induced by tail dependence. 23

1 Introduction

1.1 Motivation

Answering causal questions is a primary objective in numerous scientific disciplines. While Randomized Controlled Trials (RCTs) are considered the gold standard for analyzing causal effects, their implementation is not always applicable due to ethical, practical or cost-related constraints. Therefore, researchers often rely on observational datasets to infer causal relationships. This approach introduces a significant challenge: the potential for unobserved confounders. This occurs when an unmeasured variable influences both the treatment (T) and the outcome (Y), creating a non-causal, spurious association between T and Y that can lead to a biased estimation of the true causal effect.

Addressing this bias is critically important, as mistaken causal conclusions can lead to significant consequences such as ineffective policy decisions, inappropriate medical treatments or inaccurate scientific claims. The main challenge in avoiding these mistakes is that the key assumption for unbiased causal inference from observational data, the absence of hidden confounding, is not testable in practice. Since unmeasured factors cannot be observed, their absence cannot be empirically confirmed.

To manage this uncertainty, the field of sensitivity analysis has evolved as a means to measure how possible unobserved confounders may affect the estimation of the causal effect. Particularly, deep-learning based methodologies have allowed researchers to create non-parametric methodologies that are well-suited to this type of problem. For instance, Causal Graphical Normalizing Flows (cGNFs) [1] can be used to assess the causal relationship between two variables. One of the most recent and advanced methodologies that exemplifies this methodology is the ρ-GNF [2], which uses a Gaussian Copula to represent the distribution of unobserved confounders and only one sensitivity factor, ρ. The ρ-GNF allows the researcher to provide a clear understanding of how the estimated causal effect will vary depending upon the researcher's assumption regarding the amount of unobserved confounding.

This situation represents the most important knowledge gap that this thesis aims to address. Although the ρ-GNF is a very powerful approach, it relies on a strong and equally untestable assumption: That the latent dependency relationship between the variables is a Gaussian Copula. The existing literature relies on this assumption for its mathematical convenience but what if the true underlying dependency structure is not Gaussian? Relying on

a misspecified assumption could lead to a misleading sensitivity analysis, potentially creating a false sense of confidence in a biased result. Therefore, it is crucial to investigate the robustness of the ρ-GNF approach and to understand the consequences of violating this fundamental assumption.

1.2 Aim and Research Question

The aim of this thesis is to enhance the reliability of causal estimates derived from observational studies. The focus is to investigate the impact of the untestable latent distribution assumption on causal estimates. This investigation is critical as this assumption underlies certain modern sensitivity analysis models, for instance, the ρ-GNF. This study will provide empirical demonstrations of both the type and magnitude of deviation that occurs in the causal effect estimates of researchers in causal inference when the data violate the standard Gaussian assumption of the ρ-GNF approach. Ultimately, this research will provide the causal inference researcher with a practical demonstration of the possible limitations of their model assumptions and will also aid in developing a framework for conducting more robust sensitivity analyses.

This thesis seeks to answer the following research question:

1. How sensitive is the ρ-GNF's estimation of the Average Causal Effect (ACE) to a mismatch between the assumed Gaussian copula and the possibly non-Gaussian copula underlying the data?

2 Theoretical Background and Related Work

2.1 Theoretical Background

2.1.1 Causal Effects and the Problem of Confounding

The objective of causal inference is to measure how the treatment or cause (denoted as T) impacts an outcome or effect (denoted as Y). Assume that T represents a binary treatment variable. Therefore, when $T = 1$ it denotes an individual in the treated group while $T = 0$ denotes an individual in the untreated group. Each member of a sample has two possible outcomes:

- The potential outcome for an individual receiving the treatment (denoted as $Y(1)$)
- The potential outcome for an individual not receiving the treatment (denoted as $Y(0)$)

The causal effect for an individual is the difference between the two possible outcomes; i.e. the causal effect is $Y(1) - Y(0)$. However, it is impossible to measure the causal effect of an individual because we cannot see both possible outcomes for the same individual at the same time. Instead, we only see the actual outcome (Y) based on whether the individual was assigned to receive the treatment (T):

$$Y = T \cdot Y(1) + (1 - T) \cdot Y(0) \tag{2.1}$$

Since causal effects are not measurable for individuals, researchers usually seek to determine the Average Causal Effect (ACE) [4, 6, 7] which is the average of the differences of the potential outcomes across the entire population:

$$\text{ACE} = E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)] \tag{2.2}$$

Randomized Controlled Trials (RCTs) allow the researcher to assign the treatment (T) randomly to individuals. Therefore, randomization ensures that the treatment is independent of the possible outcomes ($Y(1), Y(0)$). With this independence, combined with the assumption of consistency, which implies that the observed outcome Y corresponds to the potential outcome $Y(t)$ when $T = t$, the expected value of the possible outcomes is identified:

$$E[Y(t)] = E[Y(t) \mid T = t] = E[Y \mid T = t] \tag{2.3}$$

However, in observational studies, the researcher does not assign the treatment (T). As a result, the treatment (T) and the outcome (Y) could have common causes (a confounder (U)), which create a "backdoor path" ($T \leftarrow U \rightarrow Y$) that results in a spurious relationship between T and Y. A failure to adjust for the confounders in observational studies will produce a biased estimate of the ACE when calculating the difference in means.

A standard requirement for identifying the ACE from observational data is the unconfoundedness assumption, which assumes that given a set of covariates (X), the treatment (T) and the potential outcomes ($Y(1), Y(0)$) are independent:

$$(Y(1), Y(0)) \perp T \mid X \tag{2.4}$$

If all of the confounders are observable (i.e., included in X), then the ACE can be identified via the adjustment formula:

$$E[Y(t)] = E_X[E[Y \mid T = t, X]] = \sum_x E[Y \mid T = t, X = x]P(X = x) \tag{2.5}$$

However, if there are unobserved confounders, variables that affect both T and Y but are not recorded in the data, the unconfoundedness assumption is violated, and the standard adjustment methods will not produce the correct causal effect. This limitation makes it necessary to apply sensitivity analysis techniques.

2.1.2 Structural Causal Models (SCMs)

The fundamental framework for conceptualizing causal relationships from observational data is provided by Structural Causal Models (SCMs)[6]. An SCM defines the causal relationships among variables in a system through a set of functional assignments. These relationships are typically visualized using a Directed Acyclic Graph (DAG).

The fundamental causal structure considered in this thesis involves a treatment variable T affecting an outcome variable Y. However, as visualized later in Figure 2.1, this relationship is not isolated; it is confounded by correlated unobserved latent variables.

This relationship is formalized in the form of structural equations:

$$T = f_T(\varepsilon_T) \tag{2.6}$$

$$Y = f_Y(T, \varepsilon_Y) \tag{2.7}$$

where f_T and f_Y represent unknown, arbitrary functions.

The terms ε_T and ε_Y are the unobserved latent variables. They represent all the unmeasured background factors that influence T and Y, respectively.

The core methodological challenge for causal inference from observational data is the existence of an unobserved confounder (U) that influences both T and Y.

In the SCM, defined by the equations 2.6 and 2.7, this confounding generates a statistical dependence between the latent variables ε_T and ε_Y.

The main objective of this study is to model the pattern of this latent association and examine it through sensitivity analysis.

2.1.3 Normalizing Flows

Normalizing Flows (NFs) operate by transforming a random variable Z from a simple base distribution (such as a normal distribution) into a variable X following a complex target distribution. This is achieved by applying a sequence of invertible and differentiable transformations f, such that $X = f(Z)$ [5].

NFs allow for the computation of the exact density of the variable X after transformation of the base distribution $q(z)$ to the target distribution $p(x)$ using the change of variables formula. The change of variables formula calculates the density of the transformed variable by

multiplying the base density by the absolute value of the determinant of the Jacobian which represents how much the volume has changed due to the transformation.

Since the model's parameters are trainable via Maximum Likelihood Estimation (MLE), they can be estimated directly without having to use the approximate inference techniques that many other generative models require.

In this thesis, we utilize Rational Quadratic Splines (RQS) as the transformation functions. We chose RQS due to their high degree of flexibility, allowing the model to capture complex distributions.

An RQS transformation operates element-wise and is built using a process where the input space is restricted to a bounding box and then this space is divided into K intervals, also known as "bins". The boundaries of these bins are defined by parameters called "knots".

Within each bin, the transformation function is defined as a monotonically increasing rational quadratic function. This structure ensures that the transformation is fully differentiable and analytically invertible which are essential requirements for flow-based models. By adjusting the coordinates and derivatives of the knots during training, the model can learn highly non-linear shapes to match the data distribution.

The model parameters θ (including the knot locations and derivatives) are optimized by minimizing the negative log-likelihood of the observed data X. The objective function is derived from the change of variables formula:

$$\mathcal{L}(\theta) = -\mathbb{E}_{x \sim p_{\text{data}}} \left[\log p_Z(f_\theta^{-1}(x)) + \log \left| \det \frac{\partial f_\theta^{-1}(x)}{\partial x} \right| \right] \tag{2.8}$$

where p_Z is the base distribution (typically a standard Gaussian) and the second term is the log-determinant of the Jacobian, which accounts for the change in volume induced by the transformation.

2.1.4 Causal Graphical Normalizing Flows

Causal Graphical Normalizing Flows extend standard Normalizing Flows (NFs) by incorporating a causal structure, represented by a Directed Acyclic Graph (DAG) [1].

The validity of this approach relies on the product decomposition of the joint probability density. The joint distribution over all variables can be represented using the chain rule of probability in DAGs [7], so that the distribution is decomposed into the product of simpler conditional densities. Thus, instead of representing the whole system together, cGNF represents each variable with respect to its parents (causes) alone in the graph.

The decomposition of the joint probability distribution over the observed variables T and Y enables the representation of $p(T, Y)$ as follows

$$p(T, Y) = p(Y \mid T) \cdot p(T) \tag{2.9}$$

By utilizing this structure, the cGNF will learn the flow separately for each factor ($p(T)$ and $p(Y \mid T)$). The joint distribution is then constructed by the product of these conditional estimates. The cGNF will ensure that the generative model captures the causal dependencies and independencies defined by the DAG.

2.1.5 Copulas

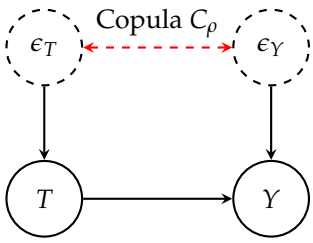


Figure 2.1: DAG illustrating the causal structure with copula-based confounding. The treatment T and outcome Y are influenced by their respective unobserved latent terms ϵ_T and ϵ_Y. The dependency structure between these latent terms is modeled by the copula C_ρ, representing the unobserved confounding.

The Sensitivity Analysis central to this thesis is based on assumptions about latent dependency structure between ε_T and ε_Y. The aim of Sensitivity Analysis is to assess how robust the estimated Average Causal Effect (ACE) is under different levels of dependency between these latent variables.

As illustrated in Figure 2.1, this dependency is represented using copulas. Sklar's theorem[9] states that any bivariate distribution function $H(x, y)$ can be expressed in terms of the individual marginal distribution functions $F(x)$ and $G(y)$ and a copula function C representing the dependency between $F(x)$ and $G(y)$:

$$H(x, y) = C(F(x), G(y)) \tag{2.10}$$

In this thesis, we focus on the use of Gaussian copulas to model the latent dependency structure between the unobserved confounders ε_T and ε_Y. The Gaussian copula is based on a multivariate normal distribution and can be represented as follows:

$$C_\rho(u, v) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v)) \tag{2.11}$$

where Φ_ρ represents the joint cumulative distribution function (CDF) of a bivariate standard normal distribution with correlation coefficient $\rho \in [-1, 1]$ and Φ^{-1} represents the inverse CDF (quantile function) of the standard univariate normal distribution. The Gaussian Copula allows for modeling the dependence between variables separately from the marginals.

2.2 Related Work

The sensitivity analysis literature for causal inference is long-standing, it has its roots in Cornfield's work on whether there was a causal relationship between smoking and lung cancer [3]. Since Cornfield's time, many different frameworks have been introduced to evaluate the robustness of causal estimates to potential unmeasured confounders. The most well-known are Rosenbaum's classic methods [8] where he calculated specific values (bounds) of the magnitude of a potentially unmeasured confounder that would completely explain away the observed effect.

Building on these foundations, Balgi et al. (2024) recently introduced the ρ-GNF technique, utilizing deep learning and generative models [2]. The ρ-GNF is the core methodology for this thesis. While most bounding approaches to partial identification typically produce very large and uninformative intervals, the ρ-GNF enables the identification of the causal effect as a continuous function of the sensitivity parameter. It uses a Causal Graphical Normalizing Flow framework to model the data-generating mechanism, while explicitly including a copula-based representation of the unobserved confounding variables.

The key strength of ρ-GNF is that it provides a way to reduce the complex dependence relationship among the latent confounders to a single dimension with a single scalar parameter, $\rho \in [-1, 1]$. By using a Gaussian copula relationship for the latent variables, changing ρ can be used to explore the entire space of possible confounding strengths. That is, the degree of confounding represented by ρ represents all of the relationships between the latent terms for the treatment and the outcome variable.

In addition to providing a means of representing the dependency between the unknown confounders as a single parameter, ρ, this parameterization provides the capability to generate a ρ-curve. The ρ-curve is a continuous visualization of the estimated ACE as a function of the sensitivity parameter, ρ. Rather than producing a single point estimate or static interval bounds of the causal effect, the ρ-curve is a dynamic representation of how the causal conclusions will vary under different levels of assumed confounding.

As with many methodologies, ρ-GNF has several advantages, including being a powerful and interpretable sensitivity analysis technique. However, like other methodologies, ρ-GNF also has significant assumptions. In particular, the assumption of a Gaussian copula relationship for the latent dependencies is a critical one. Currently, there is no systematic evaluation in the literature regarding the robustness of the ρ-curves generated from ρ-GNF against misspecifications where the true latent confounding structure is non-Gaussian. Therefore, the purpose of this thesis is to provide such an evaluation.

3 Method

3.1 Experimental Design Strategy

To empirically investigate the robustness of the ρ-GNF estimator against misspecified latent assumptions, this thesis adopts a controlled simulation-based strategy. The reason for utilizing synthetic data rather than real-world observational data is the necessity of access to the ground truth. In real-world scenarios, the true ACE and the true underlying distribution of unobserved confounders are unknown. Consequently, it is impossible to definitively measure how much a model's estimate deviates from reality due to assumption violations.

Therefore, by generating synthetic datasets where the DGP, the true causal effect and the true structure of latent variables are completely under control and known, we are able to quantify the estimation bias and test the robustness of the ρ-GNF model. The overall experimental design of our study will follow a two-step process: baseline evaluation and stress testing.

3.1.1 Baseline Assessment

This initial phase of the study validates the quality of the ρ-GNF implementation by evaluating how well the model recovers the true ACE under perfect alignment between the model specifications and the characteristics of the generated data. Specifically, in this phase, the latent variables (ε_T, ε_Y) are generated from a standard multivariate normal distribution with a known correlation coefficient.

Therefore, the purpose of the baseline assessment is to evaluate whether the ρ-GNF will be able to correctly identify the true ACE given the model's specification exactly matches the conditions under which the synthetic data were generated. Thus, it provides a benchmark against which later phases can be compared.

3.1.2 Stress Testing under Assumption Violation

The second phase of the study involves systematically violating the Gaussian copula assumption of the ρ-GNF that the latent variables are Gaussian copula distributed. This is done to evaluate the model's performance under such misspecifications. We generate data from non-Gaussian latent structures (e.g., quadratic latent variable models that have non-linear

relationships among the variables or Clayton copula latent variable models that exhibit tail dependency) while maintaining the same causal relationship as specified by the causal graph ($T \to Y$).

We then apply the ρ-GNF model to each of these non-Gaussian datasets. The ρ-GNF model is formulated based on a Gaussian copula, which inherently assumes a specific Gaussian dependence structure between the latent terms. Therefore, when we compare the ρ-GNF model's estimated ρ-curves to the true ACE values in each case, we can measure the stress placed upon the model. This allows us to determine the magnitude and direction of the bias introduced when the researcher incorrectly assumes a Gaussian structure for non-Gaussian confounding.

3.2 Data Generating Processes

In all experiments, we consider the same causal structure for each of the DAGs, represented by $T \to Y$ where T represents the treatment variable and Y represents the outcome variable. The functional representations and distributions of the unmeasured confounders differ between the various scenarios in order to evaluate the robustness of our model. Below are the descriptions of the SCMs used to generate the synthetic data for each scenario.

3.2.1 Gaussian Copula DGP (Baseline)

The first DGP serves as the baseline scenario where the DGP perfectly aligns with the assumptions of the ρ-GNF model. In this scenario, the relationships between all variables are linear and the latent variables have standard multivariate normal distributions.

The SCM for this Gaussian case is defined as follows:

$$T := \varepsilon_T \tag{3.1}$$

$$Y := \alpha T + \varepsilon_Y \tag{3.2}$$

where α represents the causal effect of T on Y.

The latent variables ε_T and ε_Y are jointly distributed as a bivariate normal distribution with correlation coefficient ρ:

$$\begin{pmatrix} \varepsilon_T \\ \varepsilon_Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right) \tag{3.3}$$

The ACE can be derived directly from the definition of the ACE. By intervening on T (setting $T = t$), the structural equation for Y becomes $Y(t) = \alpha t + \varepsilon_Y$. Thus:

$$\text{ACE} = E[Y(1) - Y(0)] = E[(\alpha \cdot 1 + \varepsilon_Y) - (\alpha \cdot 0 + \varepsilon_Y)] = E[\alpha] = \alpha \tag{3.4}$$

In this baseline scenario, the Average Causal Effect is identifiable, equal to α and known. Therefore, the dataset is used to verify whether the ρ-GNF can accurately recover the true ACE given the true ρ when the Gaussian assumptions hold.

3.2.2 Non-Gaussian DGP (Quadratic)

The second DGP serves as a stress test by introducing a misspecification where the latent dependency structure is non-linear. While the ρ-GNF assumes a Gaussian copula (which essentially models linear correlation), this DGP has a specific non-Gaussian joint distribution where the relationship between the latent variables is quadratic (U-shaped).

To ensure the non-Gaussian latent variables produced by the DGP remain standardized so that they can be used fairly for comparisons, the DGP includes three sequential components.

First, the latent variables for treatment ε_T and an independent auxiliary variable η are sampled from Standard Normal Distributions:

$$\varepsilon_T, \eta \overset{i.i.d.}{\sim} \mathcal{N}(0,1) \tag{3.5}$$

Next, the latent variables for the outcome (ε_Y) is created as a function of the latent variable for treatment (ε_T) such that it exhibits a quadratic dependence:

$$\tilde{\varepsilon_Y} = \beta\varepsilon_T^2 + \eta \tag{3.6}$$

where the strength of the quadratic effect is controlled by the parameter β.

Third, since both ε_T and the standard normal distributed terms have mean 0 and variance 1, we need to normalize $\tilde{\varepsilon_Y}$ to produce the final latent term (ε_Y):

$$\varepsilon_Y = \frac{\tilde{\varepsilon_Y} - \mu_{\tilde{\varepsilon_Y}}}{\sigma_{\tilde{\varepsilon_Y}}} \tag{3.7}$$

where $\mu_{\tilde{\varepsilon_Y}}$ and $\sigma_{\tilde{\varepsilon_Y}}$ denote the mean and standard deviation of the variable $\tilde{\varepsilon_Y}$, respectively. This standardization step ensures that the final latent variable ε_Y has zero mean and unit variance.

Lastly, the observed variables T and Y are constructed via the structural equations:

$$T := \varepsilon_T \tag{3.8}$$

$$Y := \alpha T + \varepsilon_Y \tag{3.9}$$

This generates a dataset where the marginal distributions of the latent variables are standardized to have zero mean and unit variance but the joint dependence of these latent variable is strictly non-Gaussian due to the quadratic dependency.

3.2.3 Archimedean DGP (Clayton Copula)

The third and final scenario introduces a specific form of model misspecification: Tail Dependence. The ρ-GNF uses the Gaussian copula which has a zero tail dependence. The Clayton Copula is one of the several Archimedean copulas that is known for exhibiting strong lower tail dependence. Therefore, extreme low values of the treatment variable, ε_T, have a strong correlation with extreme low values of the latent variable for the outcome, ε_Y, i.e., the structure the Gaussian copula can't model.

To generate data with this specific dependency structure while still having standard Normal marginals, we make use of the Conditional Sampling Method. The generation process can be described as follows:

First, generate two independent random sources U_1 and V, each from a standard Uniform distribution on the interval $[0,1]$:

$$U_1, V \sim \mathcal{U}[0,1] \tag{3.10}$$

Next, generate a dependent uniform variable U_2 by inverting the conditional distribution function of the Clayton copula. This formula transforms the independent variable V into a variable that depends on U_1 according to the Clayton parameter θ:

$$U_2 = \left[U_1^{-\theta}(V^{-\frac{\theta}{1+\theta}} - 1) + 1\right]^{-\frac{1}{\theta}} \tag{3.11}$$

where $\theta > 0$ controls the strength of the dependency. A higher value of θ corresponds to a larger degree of dependence, especially in the lower tail.

Third, transform these uniform variables (U_1, U_2) into latent variables ($\varepsilon_T, \varepsilon_Y$) that follow a Standard Normal distribution using the inverse cumulative distribution function (probit function), denoted as Φ^{-1}:

$$\varepsilon_T = \Phi^{-1}(U_1) \tag{3.12}$$

$$\varepsilon_Y = \Phi^{-1}(U_2) \tag{3.13}$$

To isolate the impact of the misspecified dependency structure and ensure that the marginal distributions remain consistent with the ρ-GNF's assumption of standard normality, we standardize the raw outcome latent variable ε_Y:

$$\varepsilon_Y = \frac{\varepsilon_Y - \mu_{\varepsilon_Y}}{\sigma_{\varepsilon_Y}} \tag{3.14}$$

where μ_{ε_Y} and σ_{ε_Y} are the sample mean and standard deviation of the generated variable. Finally, the observable variables T and Y are constructed via the structural equations:

$$T := \varepsilon_T \tag{3.15}$$

$$Y := \alpha T + \varepsilon_Y \tag{3.16}$$

Thus, this process produces a dataset, where the marginal distributions have been standardized to standard normal, however the joint dependency structure has very strong lower tail dependence.

3.3 Implementation and Evaluation

For estimating the causal effects and generating the sensitivity curves (ρ-curves), we used a ρ-GNF model which was implemented with the use of Causal Graphical Normalizing Flows. The implementation uses Rational Quadratic Splines (RQS) as the base transformation function for their ability to capture complex densities well.

3.3.1 Model Architecture and Optimization

For each dataset, the training process requires learning two separate invertible transformations to learn from the data:

1. A marginal flow for the treatment variable T, denoted as f_T.
2. A conditional flow for the outcome variable Y given T, denoted as $f_{Y|T}$.

The model parameters are optimized using Maximum Likelihood Estimation (MLE). However, to perform sensitivity analysis, the training is performed repeatedly over a grid of assumed sensitivity parameters, $\rho_{assumed} \in [-0.9, 0.9]$. For each fixed $\rho_{assumed}$, the loss function is defined as the negative log-likelihood of the data, assuming that the latent variables are strictly bivariate normal with correlation $\rho_{assumed}$. The model was trained using the adam optimizer.

Hyperparameters were set to the same value for all experiments as follows:

- **Learning Rate:** 3×10^{-3}
- **Training Steps:** 1000 steps per ρ assumption.
- **Spline Bound:** 10.0 (defining the range of the spline transformation).
- **ACE Samples:** $N = 100,000$ (Monte Carlo integration sample size).

3.3.2 ACE Estimation via Monte Carlo Simulation

For a fixed $\rho_{assumed}$, the model is first trained to convergence. Subsequently, the ACE is estimated via Monte Carlo integration. We draw a large number of samples ($N = 100,000$) from the assumed latent distribution and propagate them through the trained network to ensure statistical reliability.

The estimation process for the ACE is conducted as follows:

1. Sample latent variables $(\varepsilon_T, \varepsilon_Y)$ from the assumed Bivariate Normal distribution $\mathcal{N}(0, \Sigma_\rho)$.
2. Compute the potential outcomes by passing the sampled latent variables and the fixed treatment values ($T = 1$ and $T = 0$) through the inverse of the learned transformations:

$$Y_{do(1)} = f_{Y|T}(1, \varepsilon_Y) \tag{3.17}$$

$$Y_{do(0)} = f_{Y|T}(0, \varepsilon_Y) \tag{3.18}$$

3. The ACE is calculated as the mean of the differences between the two previously calculated potential outcomes.

$$\widehat{\text{ACE}}_\rho = \frac{1}{N} \sum_{i=1}^{N} (Y_{do(1)}^{(i)} - Y_{do(0)}^{(i)}) \tag{3.19}$$

3.3.3 Experimental Setup and Stability Analysis

To provide a solid assessment of model performance and robustness, we used a simulation design with independent replications for both of the sub-experiments in our study for each of the DGPs.

General Settings

All experiments were run **20 independent runs** ($N_{runs} = 20$) to ensure simulation variability would be accounted for. For the sensitivity analysis, the assumptions regarding the parameters were tested on a grid of values: $\rho_{assumed} \in [-0.9, 0.9]$ with 19 equally spaced values.

Experiment A: Sensitivity to Causal Effect Magnitude

The first set of experiments investigated whether the model would perform similarly across varying magnitudes and signs of the true causal effect. The true causal effect coefficients are varied while keeping the noise parameters constant:

- **Fixed Noise Parameters:** $\rho = 0.7$ (Gaussian), $\beta = 0.5$ (Quadratic), $\theta = 2.0$ (Clayton).
- **Varying True ACE (α):** We test three distinct values: $\alpha \in \{-2.0, 1.0, 2.0\}$. This covers strong negative, moderate positive, and strong positive causal effects.

Experiment B: Sensitivity to Dependency Strength

The second set of experiments examine how the model performs when faced with increased levels of latent dependencies (i.e., confounding strength). In our experimental conditions, we fixed the true causal effect at $\alpha = 1.0$ and varied the parameter which controls the latent structure:

- **Gaussian Copula DGP:** We test $\rho_{true} \in \{-0.7, 0.0, 0.7\}$ to evaluate performance under negative, zero, and positive confounding.

- **Non-Gaussian (Quadratic) DGP:** We test non-linear strength $\beta \in \{-0.5, 0.0, 0.5\}$. Note that $\beta = 0$ corresponds to the linear Gaussian case, serving as a control.
- **Archimedean (Clayton Copula) DGP:** We test tail dependence parameters $\theta \in \{0.1, 2.0, 5.0\}$. Higher θ values indicate stronger lower tail dependence.

3.3.4 Evaluation Strategy

Due to the fundamental difference in the alignment between the model assumptions and the true data generating process across the scenarios, we will utilize two unique evaluation strategies:

1. Baseline Evaluation (Gaussian Consistency):

In the Gaussian Copula DGP, a true ρ exists. For this reason, we will evaluate the model's **precision**. Successful recovery of the ground truth by the model requires that the model's ρ-curve be able to intersect the true ACE line exactly at the true ρ value used to create the simulated data.

2. Stress Test Evaluation (Robustness to Misspecification):

In the Non-Gaussian and Archimedean DGPs, there is no true ρ that corresponds to the latent structure. Therefore, unlike the baseline scenario where we test for exact precision, here we evaluate the model's capacity to find a valid approximation.

Finding an intersection with the true ACE line is the primary requirement, as it indicates a mathematical solution exists. It is however not enough by itself to demonstrate robustness. Therefore, we will evaluate the model's **robustness** to misspecification using these criteria:

- **Intersection Existence:** The primary requirement is whether the estimated ρ-curve intersects the True ACE line at any point within the valid range of $\rho \in [-0.9, 0.9]$. This confirms that the ρ-GNF is flexible enough to compensate for the misspecification bias.
- **Stability (Variance):** We analyze the width of the shaded error bands, which represent the standard deviation across the 20 independent runs. Narrow error bands indicate the estimator is stable and can produce similar results under violations of the distributional assumptions. Wider error bands may indicate the misspecification significantly increases the variability of the optimization process.
- **Monotonicity:** We assess the shape of the ρ-curve. A desirable curve should be *monotonic* (smoothly increasing or decreasing without erratic fluctuations). This property is crucial because it implies a consistent relationship between the assumed level of confounding and the estimated causal effect. Furthermore, monotonicity guarantees the **uniqueness** of the intersection point, ensuring that the model suggests a single "ρ" rather than multiple ambiguous solutions.

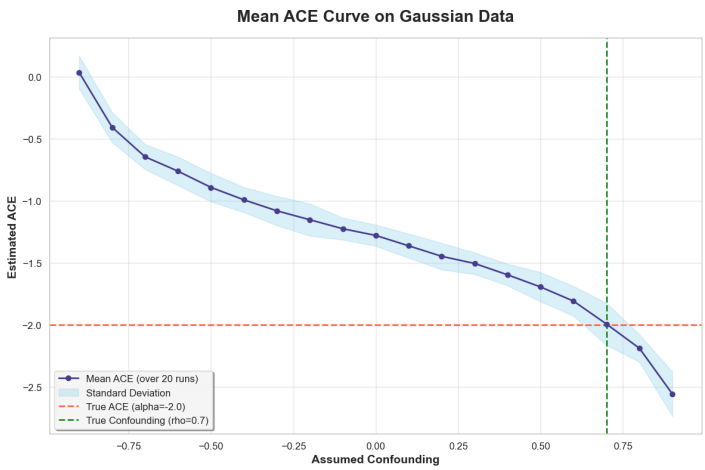
4 Results

4.1 Gaussian Copula DGP

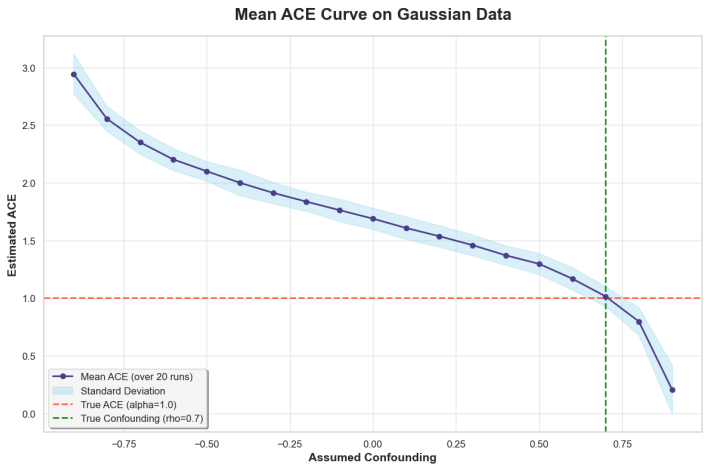
This section presents the results of the baseline assessment where the synthetic data were generated using a Gaussian copula dependency structure. In these experiments, the data generation process aligns with the structural assumptions of the ρ-GNF model.

Estimated ρ-curves and their intersections with the ground truth ACE lines (with respect to the true correlation value ρ_{true}) are presented. The results are divided into two subsections: first, observing the model's output across varying causal effect magnitudes (α), and second, observing the output across varying levels of confounding strength (ρ).

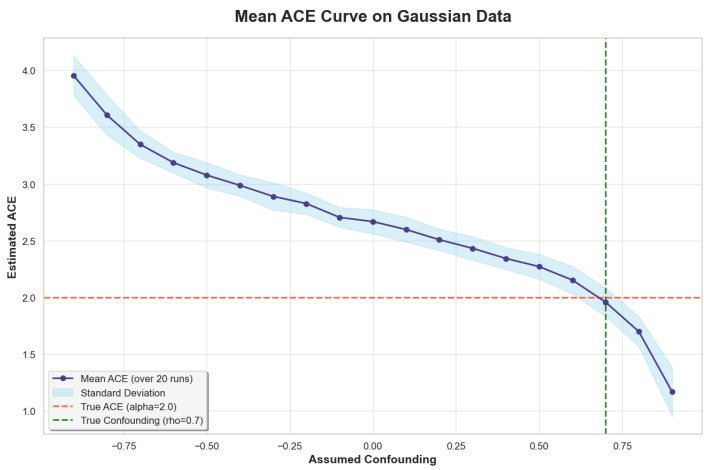
4.1.1 Sensitivity to Causal Effect Magnitude (α)



(a) $\alpha = -2.0$



(b) $\alpha = 1.0$



(c) $\alpha = 2.0$

Figure 4.1: Estimated ρ-curves for the Gaussian DGP (Baseline) with fixed true correlation $\rho_{true} = 0.7$. The graphs show the true causal effect as follows: (a) $\alpha = -2.0$, (b) $\alpha = 1.0$, and (c) $\alpha = 2.0$. The blue curve is the mean estimated ACE over 20 independent runs and the shaded region indicates the standard deviation. The horizontal red dashed line indicates the ground truth ACE while the vertical green dashed line indicates the true confounding level $\rho = 0.7$.

Figure 4.1 presents the sensitivity analysis results where the latent dependency is generated as Gaussian with a fixed correlation of $\rho_{true} = 0.7$. The plots illustrate the model's estimates across different magnitudes and directions of the true causal effect (α).

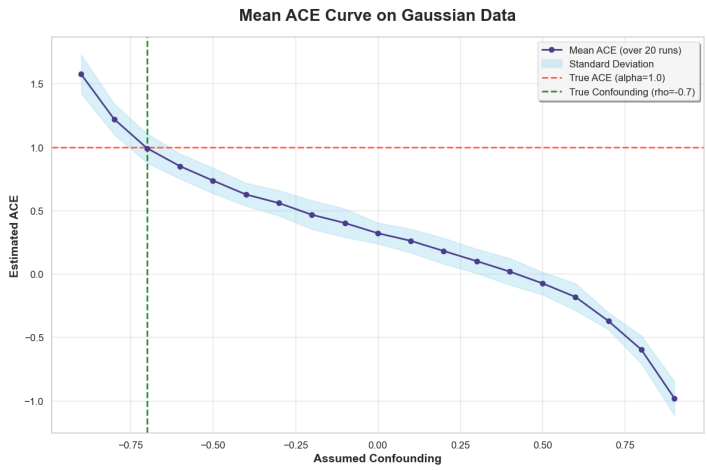
As illustrated in the figure, across all three scenarios, ranging from a strong negative effect ($\alpha = -2.0$) to a strong positive effect ($\alpha = 2.0$), the estimated ρ-curves exhibit a monotonic trend. The shaded error bands remain narrow relative to the scale of the effect across the 20 independent runs.

In every graph, the estimated ρ-curve intersects the ground truth ACE line (red) exactly at the vertical marker for the true correlation parameter (green, $\rho = 0.7$). The intersection occurs consistently at the generated confounding level across the different scales of the outcome variable.

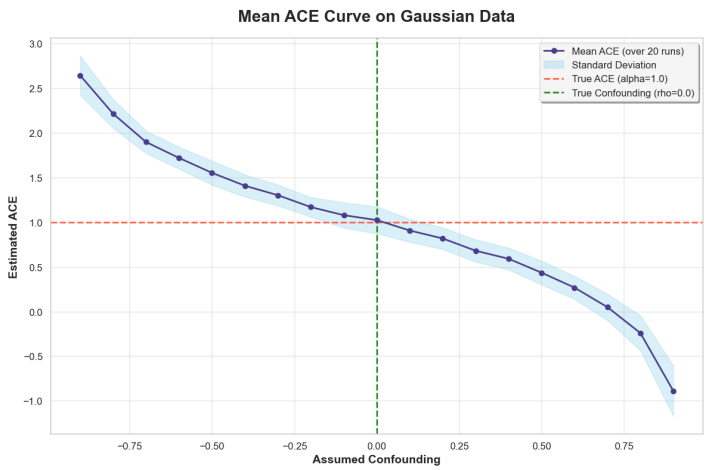
4.1.2 Sensitivity to Confounding Strength (ρ)

Figure 4.2 presents the results of the sensitivity analysis regarding confounding strength, where the latent dependency is correctly specified as Gaussian. This experiment observes the model's precision across a spectrum of confounding levels (ρ) while keeping the causal effect fixed at $\alpha = 1.0$.

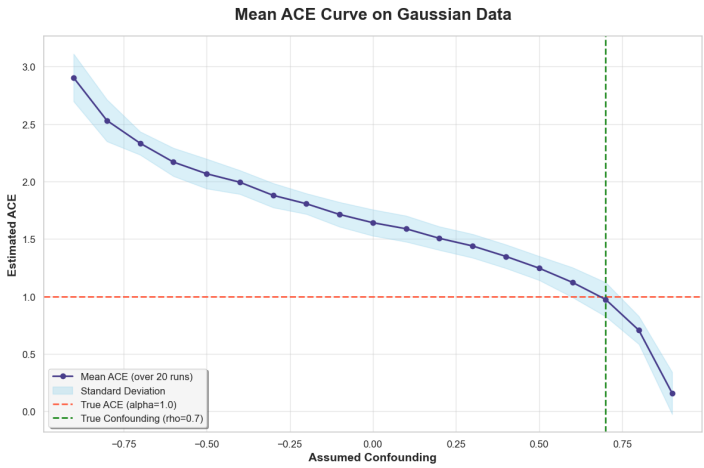
As shown in the figure 4.2, across all three scenarios, ranging from strong negative confounding ($\rho = -0.7$) to no confounding ($\rho = 0$) and strong positive confounding ($\rho = 0.7$), the estimated ρ-curves exhibit a monotonic trend. The shaded error bands remain narrow across the 20 independent runs.



(a) $\rho = -0.7$



(b) $\rho = 0$



(c) $\rho = 0.7$

Figure 4.2: Estimated ρ-curves for the Gaussian DGP (Baseline) under varying confounding strengths. The graphs show the different true confounding levels as follows: (a) $\rho = -0.7$, (b) $\rho = 0$, and (c) $\rho = 0.7$. In all cases, the true causal effect is fixed at $\alpha = 1.0$. The blue curve represent the mean estimated ACE over 20 independent runs and the shaded region indicates the standard deviation. The horizontal red dashed line marks the ground truth ACE while the vertical green dashed line marks the respective true confounding level (ρ_{true})

In each graph, the estimated ρ-curve intersects the ground truth ACE line (red) at the location of the vertical marker for the true correlation parameter (green). The intersection point shifts consistently in a harmony with the generated confounding level (ρ_{true}) across the different scenarios.

4.2 Non-Gaussian DGP

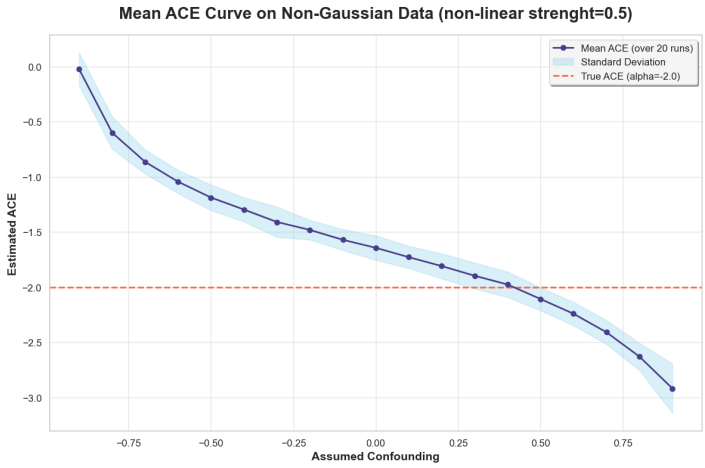
This section presents the results of the first stress test, designed to evaluate the behavior of the ρ-GNF estimator under non-linear confounding. In this scenario, the synthetic data were generated using a SCM with a quadratic dependency structure ($\varepsilon_Y \propto \varepsilon_T^2$) while the ρ-GNF model continues to assume a Gaussian copula structure.

Unlike the baseline scenario, there is no single ground truth correlation parameter (ρ_{true}) that corresponds to the quadratic structure. We investigate whether the ρ-GNF can determine an "effective" ρ value that aligns the estimated ACE with the ground truth ACE. Consequently, the analysis centers on the existence and location of an intersection between the estimated ρ-curve and the ground truth ACE line within the valid range of $\rho \in [-0.9, 0.9]$.

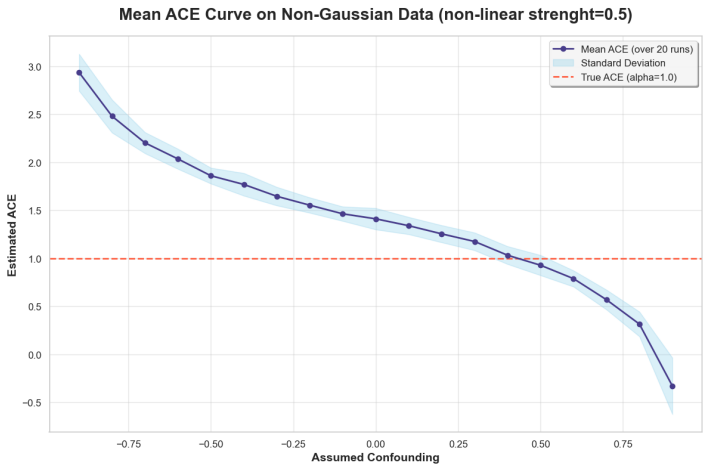
The analysis is divided into two parts: first, we examine the model's performance across different causal effect magnitudes (α) under moderate quadratic confounding, and second, we observe the estimates while varying the strength of the non-linear dependency (β).

4.2.1 Sensitivity to Causal Effect Magnitude (α)

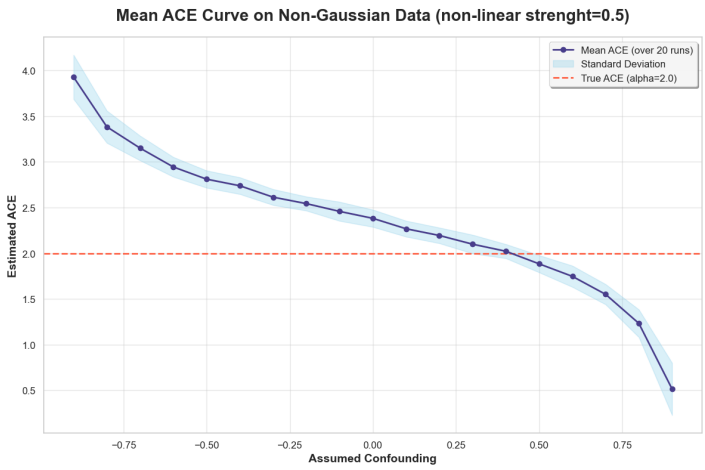
In this first set of stress tests, we fix the strength of the quadratic confounding at a moderate level ($\beta = 0.5$) and vary the true ACE ($\alpha \in \{-2.0, 1.0, 2.0\}$). Since the data generating process is non-Gaussian, there is no "true ρ" to serve as a reference. Instead, the analysis examines whether the estimated ρ-curve intersects the true ACE line within the valid range.



(a) $\alpha = -2.0$



(b) $\alpha = 1.0$



(c) $\alpha = 2.0$

Figure 4.3: Estimated ρ-curves for the Quadratic scenario ($\beta = 0.5$). The graphs correspond to true causal effects: (a) $\alpha = -2.0$, (b) $\alpha = 1.0$, and (c) $\alpha = 2.0$. Since the true dependency is non-Gaussian, there is no vertical marker for ρ_{true}. The intersection of the blue curve with the red true ACE line indicates the "effective ρ" value where the estimate aligns with the ground truth.

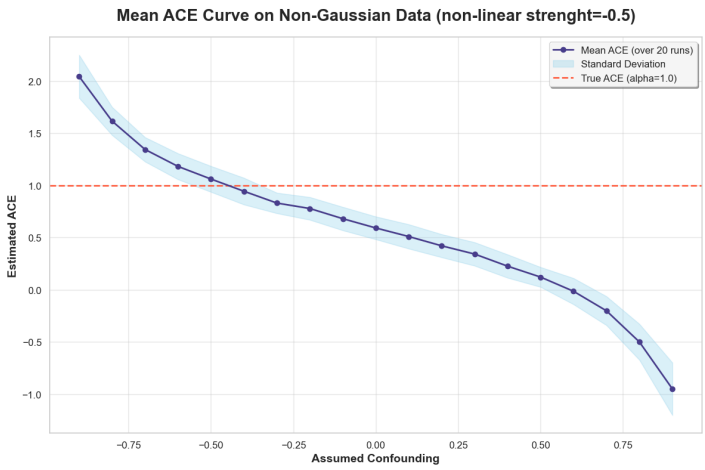
In the Figure 4.3, despite the structural mismatch between the quadratic DGP and the Gaussian model assumption, in all three scenarios, the estimated ρ-curve intersects the ground truth ACE line within the valid range of ρ.

Additionally, the width of the shaded error bands remains consistent across the different magnitudes of the causal effect, indicating that the variance of the estimator is not sensitive to the scale of α under this specific misspecification.

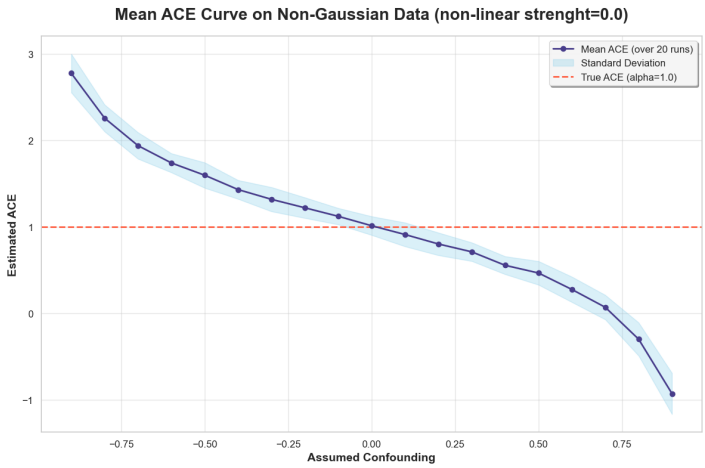
4.2.2 Sensitivity to Non-Linear Strength (β)

In the second phase of the quadratic stress test, we investigate the impact of the strength of the non-linear distortion on the estimator. Here, the true causal effect is fixed at $\alpha = 1.0$ while the non-linear strength parameter varies ($\beta \in \{-0.5, 0.0, 0.5\}$).

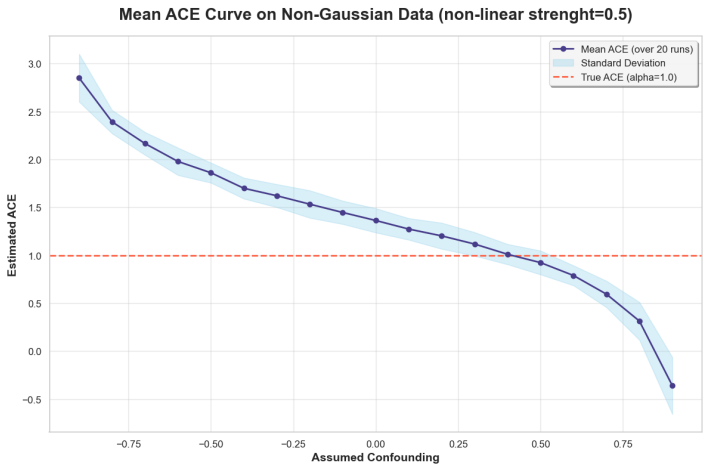
It is important to note that the case where $\beta = 0.0$ serves as a **linear control scenario**. When $\beta = 0$, the quadratic term vanishes and the latent variables become independent standard normal variables (equivalent to a Gaussian copula with $\rho = 0$). The other two scenarios ($\beta = -0.5$ and $\beta = 0.5$) represent the misspecified conditions with strong U-shaped dependencies.



(a) $\beta = -0.5$ (Negative Quadratic)



(b) $\beta = 0.0$ (Linear Control)



(c) $\beta = 0.5$ (Positive Quadratic)

Figure 4.4: Estimated ρ-curves for the Quadratic scenario with fixed $\alpha = 1.0$ and varying non-linear strengths. (b) Represents the control case where the dependency is linear ($\beta = 0$). (a) and (c) represent misspecified quadratic dependencies. The intersection of the blue curve with the red true ACE line denotes the effective sensitivity parameter.

Figure 4.4b displays the linear control case ($\beta = 0$), where the DGP effectively reverts to a Gaussian structure with independent noise. As observed, the estimated ρ-curve intersects the ground truth ACE line precisely at $\rho = 0$, consistent with the lack of confounding.

In Figures 4.4a and 4.4c, where $\beta = -0.5$ and $\beta = 0.5$, the latent dependency structure is strictly non-Gaussian. Despite this violation, the estimated ρ-curves in both cases intersect the ground truth ACE line within the valid range. Notably, the location of this intersection shifts according to the sign of the distortion parameter: for the negative quadratic strength ($\beta = -0.5$), the intersection occurs in the negative ρ domain, while for the positive quadratic strength ($\beta = 0.5$), the intersection shifts to the positive ρ domain.

4.3 Clayton Copula DGP

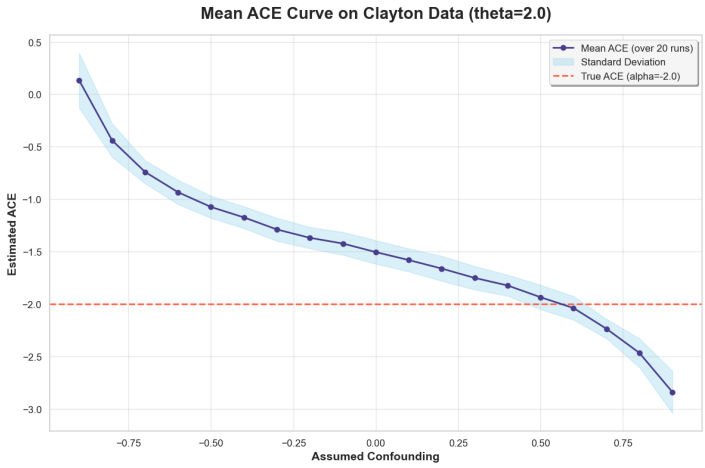
This section presents the results of the second stress test, where the synthetic data were generated using an Clayton (Archimedean) copula. This scenario introduces a specific type of structural mismatch known as **tail dependence**. While the ρ-GNF model assumes a Gaussian dependence structure (characterized by asymptotic independence in the tails), the generated data exhibits strong dependency specifically in the lower tail of the distribution.

Similar to the quadratic scenario, there is no single ground truth correlation parameter (ρ_{true}) that corresponds directly to the Clayton dependence parameter (θ). We report whether the estimated ρ-curves intersect the ground truth ACE line within the valid range of the sensitivity parameter $\rho \in [-1, 1]$.

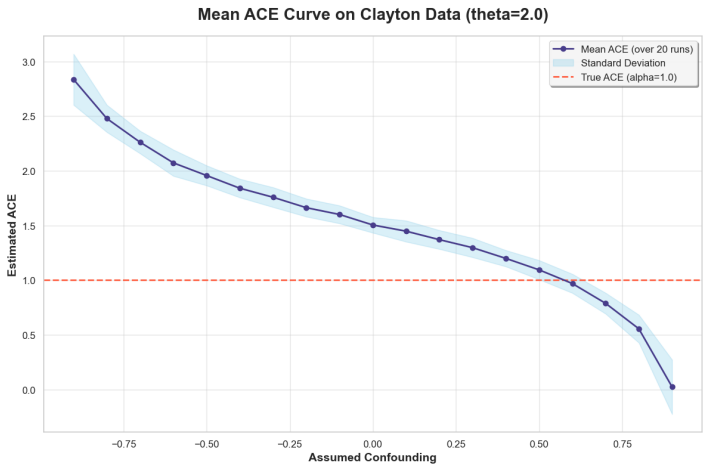
The results are organized into two subsections: first, observing the model's estimates across varying causal effect magnitudes (α) under a fixed level of tail dependence, and second, observing the estimates across varying strengths of the tail dependency (θ).

4.3.1 Sensitivity to Causal Effect Magnitude (α)

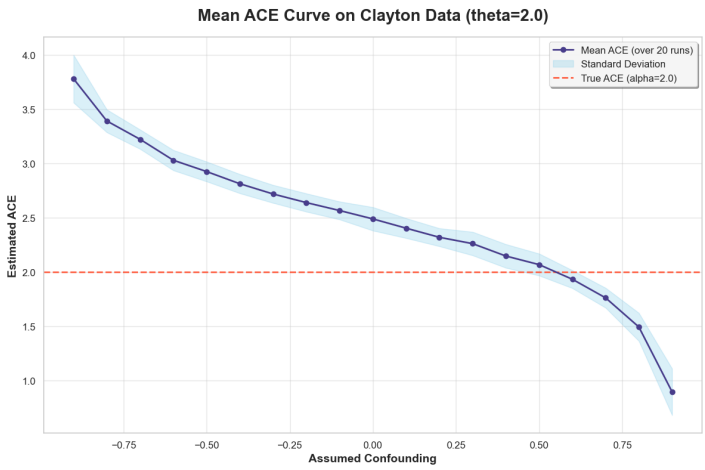
In this first experiment for the Archimedean scenario, we fix the tail dependence parameter at a moderate level ($\theta = 2.0$) and vary the true ACE ($\alpha \in \{-2.0, 1.0, 2.0\}$). Similar to the quadratic case, there is no direct mapping between the Clayton parameter θ and the Gaussian parameter ρ. Therefore, the analysis examines the existence of an intersection point between the estimated ρ-curve and the ground truth ACE line.



(a) $\alpha = -2.0$



(b) $\alpha = 1.0$



(c) $\alpha = 2.0$

Figure 4.5: Estimated ρ-curves for the Archimedean (Clayton) scenario ($\theta = 2.0$). The graphs correspond to true causal effects: (a) $\alpha = -2.0$, (b) $\alpha = 1.0$, and (c) $\alpha = 2.0$. Since the true dependency involves tail dependence (non-Gaussian), there is no vertical marker for ρ_{true}. The intersection of the blue curve with the red true ACE line denotes the effective sensitivity parameter.

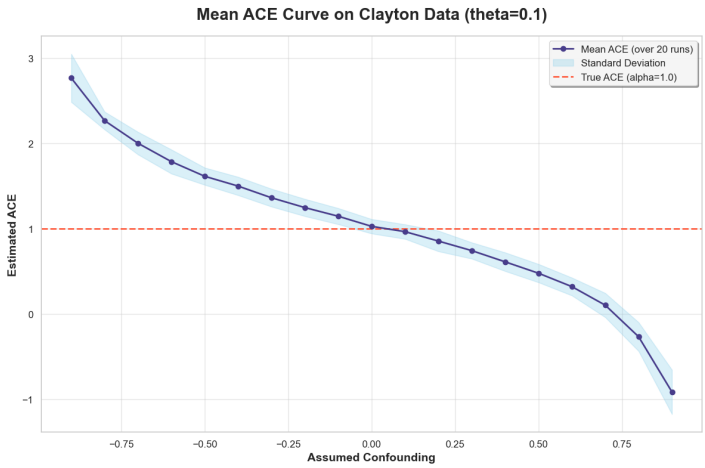
The figure 4.5 illustrate that the ρ-GNF estimator produces stable ρ-curves even when the underlying data exhibits lower tail dependence, a feature not present in the Gaussian copula.

In all three scenarios, the estimated ρ-curve intersects the ground truth ACE line within the valid range of ρ. The consistency of the error bands across the different α values indicates that the magnitude of the causal effect does not change the model's response to the tail dependence misspecification.

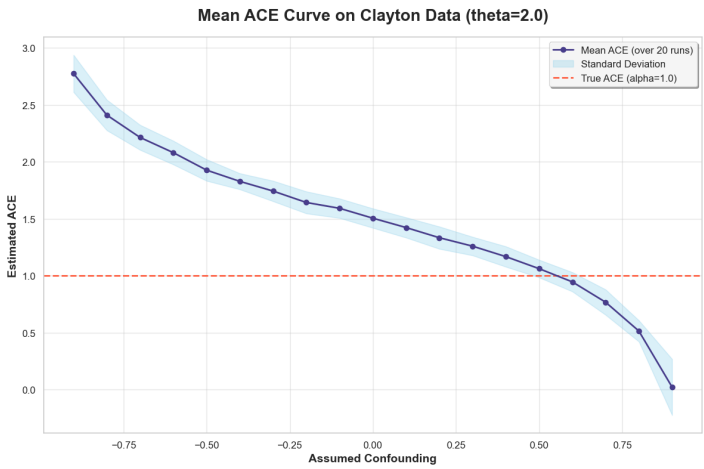
4.3.2 Sensitivity to Tail Dependence Strength (θ)

In the final phase of the stress testing, we investigate the impact of the *strength* of tail dependence on the estimator. Here, the true causal effect is fixed at $\alpha = 1.0$, while the Clayton copula parameter varies ($\theta \in \{0.1, 2.0, 5.0\}$).

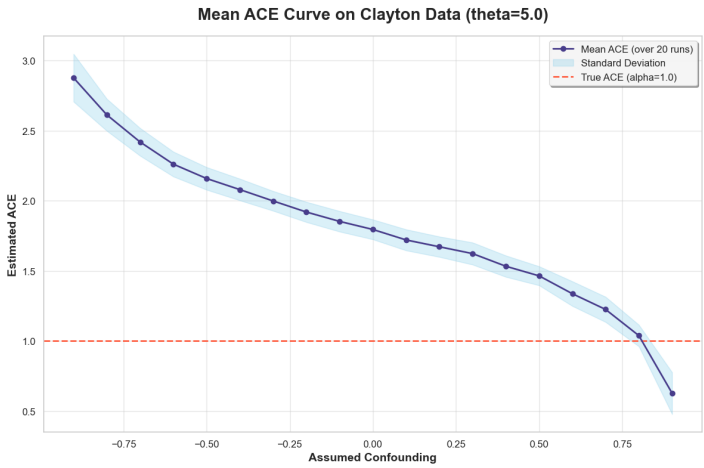
It is important to note that lower values of θ (e.g., 0.1) correspond to weak dependence, approaching the independent case. Conversely, higher values (e.g., 5.0) represent strong lower tail dependence, constituting a severe deviation from the Gaussian copula assumption.



(a) $\theta = 0.1$ (Weak Dependence)



(b) $\theta = 2.0$ (Moderate Dependence)



(c) $\theta = 5.0$ (Strong Dependence)

Figure 4.6: Estimated ρ-curves for the Archimedean (Clayton) scenario with fixed $\alpha = 1.0$ and varying tail dependence strengths. The graphs correspond to: (a) Weak dependence $\theta = 0.1$, (b) Moderate dependence $\theta = 2.0$, and (c) Strong dependence $\theta = 5.0$. The intersection of the blue curve with the red true ACE line indicates the effective sensitivity parameter required to approximate the bias induced by tail dependence.

In Figure 4.6a, where $\theta = 0.1$, the dependence is minimal. Accordingly, the estimated ρ-curve intersects the ground truth ACE line near $\rho = 0$, consistent with the lack of strong confounding.

On the other hand, in figure 4.6b and 4.6c, as the tail dependence strengthens ($\theta = 2.0$ and $\theta = 5.0$), the intersection point shifts towards higher positive values. Specifically, the intersection occurs at a higher ρ value for $\theta = 5.0$ compared to $\theta = 2.0$. Despite the structural mismatch, where the data exhibits asymmetric tail dependence while the model assumes symmetric Gaussian dependence, an intersection with the ground truth ACE line is observed in all cases within the valid range of ρ. The width of the error bands remains stable even under the strongest misspecification ($\theta = 5.0$).

5 Discussion

This chapter interprets the empirical findings that have been presented in the results chapter, evaluates the methodological choices made throughout the study and discusses the broader implications of the research. The discussion is structured as follows: an interpretation of results, a critique of the methodology, a review of ethical considerations, and suggestions for future research.

5.1 Results

The purpose of this thesis was to investigate how sensitive the ρ-GNF estimator to misspecified latent structures. The empirical results provide significant insight into both the validity of the model when the model under ideal conditions and the model's robustness when violations occur under assumptions.

5.1.1 Performance under Gaussian Assumptions

The results from the Gaussian DGP scenarios where the data generating process perfectly matches with the structural assumptions of the ρ-GNF model, demonstrate that the estimator performs well in recovering the true Average Causal Effect (ACE). The estimated ρ-curves intersect the Ground Truth ACE line exactly at the true correlation parameter (ρ_{true}), confirming the model's validity under ideal conditions.

Furthermore, the narrow width of the shaded error bands indicates that the estimator is precise and stable across repeated runs. These findings establish that the ρ-GNF correctly recovers the causal effect when the assumed copula structure reflects the true latent dependency.

5.1.2 Robustness to Misspecification

The results from the stress test scenarios (Quadratic and Clayton DGPs), where the assumption of a Gaussian copula was deliberately violated, revealed a significant degree of model flexibility.

In both the non-linear (Quadratic) and tail-dependent (Clayton) scenarios, we observed that the estimated ρ-curves consistently intersected the True ACE line within the valid pa-

rameter range ($\rho \in [-0.9, 0.9]$). This finding indicates that, despite the structural mismatch, there exists a specific value of the sensitivity parameter ρ that allows the Gaussian model to recover the true causal effect.

Crucially, the experimental results demonstrated a mechanistic pattern in how the model achieved this recovery. As detailed in the Results chapter, the point of intersection systematically shifted depending on the nature of the misspecification. For instance, in the Quadratic case, the sign of the effective ρ mirrored the direction of the non-linearity (negative β led to negative ρ, positive β to positive ρ). Similarly, in the Clayton scenarios, stronger tail dependence (θ) progressively pushed the effective ρ further into the positive domain. This suggests that the shift in ρ functions as a compensatory mechanism: the model approximates the complex, unmodeled dependencies (such as tail asymmetry) by adjusting the linear correlation parameter to a value that balances out the confounding bias.

Importantly, the stability of the estimates (indicated by narrow error bands) and the preservation of monotonicity across all stress tests suggest that the ρ-GNF optimization landscape remains robust even when applied to non-Gaussian data. The model does not show erratic behavior, suggesting that the Gaussian copula can provide a useful approximation for more complex dependency structures in the estimation of the causal effect.

It is important to observe that the model maintained this robustness regardless of the magnitude of the misspecification. Whether the tail dependence was weak ($\theta = 0.1$) or extreme ($\theta = 5.0$), the ρ-GNF consistently yielded a solution space containing the true causal effect. This indicates that the estimator's flexibility is not limited to minor violations but extends to scenarios with strong structural mismatches.

This observed compensatory mechanism allows for a broader generalization regarding the behavior of the estimator under different types of tail dependence. In the Clayton scenario, we observed that stronger lower tail dependence shifted the effective sensitivity parameter towards higher positive values. Considering that the Gaussian copula is diagonally symmetric, implying that the dependence structure in the lower tail mirrors that of the upper tail for a given correlation, we hypothesize that a DGP exhibiting upper tail dependence (such as a Gumbel copula) would similarly necessitate a higher positive effective sensitivity parameter, rather than shifting towards the negative domain. This suggests that the ρ-GNF compensates for intense local dependence in either tail by increasing the global correlation parameter, irrespective of whether the dependence is concentrated in the lower or upper quadrant.

5.2 Methodological Evaluation

This section critically evaluates the experimental design and the specific methodological choices made to answer the research question. While the results demonstrate the potential of the ρ-GNF, it is essential to contextualize these findings within the scope and limitations of the chosen simulation-based approach.

5.2.1 Reason for Synthetic Data Strategy

The decision to base the analysis entirely on on synthetic data was motivated by the critical need to establish the ground truth. In real-world observational studies, the true causal effect and the underlying nature of unobserved confounding are inherently unknown.Therefore, validating a sensitivity analysis methodology via real data is problematic. It is impossible to know if an estimate deviates from the "true" causal effect due to errors in model specification or some other source of error such as measurement error or selection bias.

By implementing controlled Data Generating Processes (DGPs), we ensured experimental control. This allowed us to isolate the effect of the Gaussian copula assumption from other potential sources of error. This isolation ensures that any deviation in the results is strictly a

consequence of the mismatch between the ρ-GNF's Gaussian assumption and the true latent structure.

5.2.2 Selection of Misspecification Scenarios

The choice of Quadratic and Clayton copula DGPs was strategic, intended to represent two distinct classes of structural violations:

- The Quadratic DGP tested the model's flexibility under *functional form misspecification*. This scenario simulated a case where the dependency is symmetric but non-linear (U-shaped), challenging the linear correlation assumption of the Gaussian copula.
- The Clayton copula DGP tested the model's flexibility under *distributional asymmetry*. This scenario introduced lower tail dependence, a feature that the Gaussian copula (which assumes asymptotic independence) is theoretically incapable of modeling.

By selecting these divergent scenarios, the study provides a comprehensive assessment of the ρ-GNF's flexibility across different types of "non-Gaussianity."

5.2.3 Limitations of the Study

While the experimental design was carefully controlled, this study is subject to several limitations:

- **Generalizability to High Dimensions:** The experiments were conducted in a simplified bivariate setting ($T \to Y$) without observed covariates (X). In practical applications, causal graphs are often high-dimensional, involving complex conditional independencies. While the core mechanism of the ρ-GNF relies on the bivariate copula between latent terms, the interaction between misspecified latent structures and high-dimensional observed covariates remains unexplored in this thesis.
- **Finite Set of Misspecifications:** We tested only two specific non-Gaussian structures (Quadratic and Clayton). While these were selected strategically to represent functional nonlinearity and distributional asymmetry, they do not capture the full spectrum of possible latent dependencies. Therefore, the robustness observed here should not be assumed to apply universally to all possible forms of non-Gaussian distributions.
- **The "True ρ" Problem in Practice:** A critical challenge highlighted by the results is the practical identification of the specific ρ value that corrects the bias. In our simulations, we could identify the intersection point because the Ground Truth ACE was known. However, in real-world empirical research, the true ACE is unknown. Consequently, while this study demonstrates that the ρ-GNF can theoretically recover the true effect under misspecification, it does not provide a method for a practitioner to pinpoint a single corrective ρ value. However, as discussed in Section 5.1.2, the estimator exhibits a systematic compensatory mechanism. Therefore, future applications could focus on identifying a plausible interval for the sensitivity parameter ρ that accounts for this mechanism, rather than seeking a singular "correct" point without strong external domain knowledge. This limits the practical applicability of the finding unless domain knowledge can guide the selection of the sensitivity parameter.

5.3 Ethical Considerations

This study relies solely on synthetic data generated through controlled probabilistic algorithms (Gaussian, Quadratic, and Clayton DGPs). Consequently, the research does not involve human subjects, personal data, or sensitive information. Therefore, standard ethical

concerns regarding informed consent, data privacy and confidentiality are not applicable to the experimental phase of this work.

5.4 Future Work

The findings of this thesis point to several promising directions for future research, aiming to further evaluate the robustness and extend the applicability of the ρ-GNF estimator.

- **Algorithmic Optimization and Statistical Power:** Due to computational constraints, this study utilized a fixed set of hyperparameters and a limited number of independent runs ($N = 20$). Future research could use hyperparameter tuning to optimize the deep learning components of the model. Increasing the number of simulation runs would improve the precision of our results by minimizing random error. This would allow us to observe the model's stability in greater detail, especially at the extreme ends of the parameter range.
- **Expanding the Simulation Scenarios:** This study focused on specific non-Gaussian structures (Quadratic and Clayton) and a bounded range of causal effects. Future work should expand this catalog to include a broader range of dependency structures. Additionally, testing the model over a wider range of causal effect magnitudes (α) and more complex functional relationships between treatment and outcome would provide a more exhaustive map of the estimator's limitations.
- **Incorporation of Covariates:** This thesis isolated the bivariate latent mechanism ($T \rightarrow Y$) to study misspecification in a controlled setting. A critical next step is to investigate how the ρ-GNF behaves when conditioned on high-dimensional observed covariates. This would involve simulating more complex causal graphs and assessing whether the robustness observed in the bivariate case extends to multivariate settings.
- **Theoretical Formalization of Robustness:** This thesis provides strong empirical evidence for the robustness of the ρ-GNF estimator via simulation studies. A critical avenue for future research is to establish a formal theoretical framework that explains this behavior. Specifically, future work could aim to derive analytical bounds on the bias introduced by misspecified copulas.

6 Conclusion

How sensitive is the ρ-GNF's estimation of the Average Causal Effect (ACE) to a mismatch between the assumed Gaussian copula and the true non-Gaussian copula underlying the data?

Based on our findings, we conclude that the ρ-GNF estimator is remarkably robust rather than sensitive to violations of the Gaussian copula assumption. The empirical findings from both non-linear (quadratic) and tail-dependent (Clayton) stress tests demonstrate that despite the violation of the Gaussian copula assumption, the model successfully recovered the true ACE. This indicates that the Gaussian copula is flexible enough to compensate for complex noise structures, as the estimated sensitivity curve consistently contains a specific bias-correcting ρ value. Consequently, the validity of the ACE estimates does not strictly require the latent noise to be Gaussian, rendering the ρ-GNF a robust tool for a much wider range of observational studies.

Use of Generative AI

Generative AI tools were utilized to assist with specific writing and technical tasks. In the context of writing, these tools were employed to refine the grammatical structure, enhance the academic tone, and identify typographical errors. Additionally, they assisted in troubleshooting LaTeX formatting issues.

Regarding the technical implementation, generative AI served as an assistive tool for debugging code and resolving engineering challenges encountered during the development of the experimental framework.

Bibliography

[1] Sourabh Balgi, Adel Daoud, Jose M. Peña, Geoffrey T. Wodtke, and Jesse Zhou. *Deep Learning With DAGs*. 2024. URL: https://arxiv.org/abs/2401.06864.

[2] Sourabh Balgi, Jose M. Peña, and Adel Daoud. "ρ-GNF: A Copula-based Sensitivity Analysis to Unobserved Confounding Using Normalizing Flows". In: *Proceedings of the 12th International Conference on Probabilistic Graphical Models (PGM)*. 2024. URL: https://proceedings.mlr.press/v246/balgi24a.html.

[3] Jerome Cornfield, William Haenszel, E. Cuyler Hammond, Abraham M. Lilienfeld, Michael B. Shimkin, and Ernst L. Wynder. "Smoking and Lung Cancer: Recent Evidence and a Discussion of Some Questions". In: *JNCI: Journal of the National Cancer Institute* (1959). DOI: 10.1093/jnci/22.1.173. URL: https://doi.org/10.1093/jnci/22.1.173.

[4] Guido W. Imbens and Donald B. Rubin. *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge University Press, 2015. URL: https://books.google.se/books?id=Bf1tBwAAQBAJ.

[5] George Papamakarios, Eric Nalisnick, Danilo Jimenez Rezende, Shakir Mohamed, and Balaji Lakshminarayanan. *Normalizing Flows for Probabilistic Modeling and Inference*. 2021. URL: https://arxiv.org/abs/1912.02762.

[6] Judea Pearl. "Causal inference in statistics: An overview". In: *Statistics Surveys* (2009). DOI: 10.1214/09-SS057. URL: https://doi.org/10.1214/09-SS057.

[7] Judea Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2009. DOI: 10.5555/1642718. URL: https://dl.acm.org/doi/book/10.5555/1642718.

[8] Paul R. Rosenbaum. "Observational Studies". In: Springer, 2002. DOI: 10.1007/978-1-4757-3692-2_1. URL: https://doi.org/10.1007/978-1-4757-3692-2_1.

[9] Abe Sklar. *Fonctions de Répartition À N Dimensions Et Leurs Marges*. Université Paris 8, 1959. URL: https://books.google.se/books?id=nreSmAEACAAJ.